

Week 1 - Introduction to AI Engineering

Concepts:

What is AI Engineering?, History of AI

Project:

Create a personal AI project idea

Outcome:

Understand the basics of AI Engineering

Week 2 - Mathematics for AI

Concepts:

Linear Algebra, Calculus, Probability

Project:

Practice linear algebra and calculus problems

Outcome:

Understand mathematical concepts for AI

Week 3 - Programming Fundamentals

Concepts:

Python basics, Data structures, Object-Oriented Programming

Project:

Build a simple AI project using Python

Outcome:

Understand programming fundamentals for AI

Week 4 - Machine Learning Fundamentals

Concepts:

Supervised Learning, Unsupervised Learning, Reinforcement Learning

Project:

Implement a supervised learning algorithm

Outcome:

Understand machine learning fundamentals

Week 5 - Deep Learning Fundamentals

Concepts:

Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks

Project:

Implement a deep learning algorithm

Outcome:

Understand deep learning fundamentals

Week 6 - Natural Language Processing

Concepts:

Text Preprocessing, Tokenization, Sentiment Analysis

Project:

Implement a natural language processing algorithm

Outcome:

Understand NLP fundamentals

Week 7 - Computer Vision

Concepts:

Image Preprocessing, Object Detection, Image Segmentation

Project:

Implement a computer vision algorithm

Outcome:

Understand computer vision fundamentals

Week 8 - Reinforcement Learning

Concepts:

Q-Learning, Deep Q-Networks, Policy Gradient Methods

Project:

Implement a reinforcement learning algorithm

Outcome:

Understand reinforcement learning fundamentals

Week 9 - Transfer Learning

Concepts:

Fine-Tuning, Feature Extraction, Domain Adaptation

Project:

Implement transfer learning in a project

Outcome:

Understand transfer learning fundamentals

Week 10 - Model Evaluation and Selection

Concepts:

Metrics, Cross-Validation, Hyperparameter Tuning

Project:

Evaluate and select a model for a project

Outcome:

Understand model evaluation and selection fundamentals

Week 11 - Deploying AI Models

Concepts:

Model Serving, API Design, Containerization

Project:

Deploy an AI model to a production environment

Outcome:

Understand deploying AI models fundamentals

Week 12 - Capstone Project

Concepts:

Integrate learned concepts, Apply AI to real-world problems

Project:

Implement a capstone project that integrates learned concepts

Outcome:

Apply AI to real-world problems and demonstrate mastery of AI concepts
